ELECTRON DELOCALIZATION

Tome 2

Atomic Basins, Localization and Delocalization Indices

Integrals of the pair density over atomic regions

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Scope and Notation

This tome integrates the pair density and the exchange-correlation density of Tome 1 over atomic basins. The integrals define atomic populations, population fluctuations, the localization index $\lambda(\mathcal{A})$ and the delocalization index $\delta(\mathcal{A},\mathcal{B})$. The last chapter defines the response of δ to a change in the number of electrons and the Dual Delocalization Descriptor.

Prerequisites. Tome 1: definitions, normalizations and units of ρ, ρ_2, γ_{xc} and h_{xc}.

Notation. The symbols $\mathcal{A}$ and $\mathcal{B}$ denote atomic basins. The same symbols denote the region of space and the label of the atom. Indices i, j label orbitals. Atomic units are used throughout.

Symbol	Quantity	Unit
$N(\mathcal{A})$	electron population of basin $\mathcal{A}$	dimensionless
$q(\mathcal{A})$	atomic charge	e
$N_2(\mathcal{A},\mathcal{B})$	basin-integrated pair density	dimensionless
$\sigma^2(N_{\mathcal{A}})$	population variance	dimensionless
$\lambda(\mathcal{A})$	localization index	dimensionless
$\delta(\mathcal{A},\mathcal{B})$	delocalization index	dimensionless
$S_{ij}(\mathcal{A})$	atomic overlap matrix element	dimensionless
n_i	natural orbital occupation number	dimensionless
$\Delta\delta(\mathcal{A},\mathcal{B})$	Dual Delocalization Descriptor	N^{-1} (inverse electron number)

1. Atomic Basins and Basin Integrals

1.1 Atomic basins

In the quantum theory of atoms in molecules (QTAIM), the gradient vector field $\nabla\rho(\mathbf{r})$ partitions space into atomic basins. A basin $\mathcal{A}$ contains one nuclear attractor and is bounded by a surface S that satisfies the zero-flux condition

$$\nabla\rho(\mathbf{r})\cdot\mathbf{n}(\mathbf{r}) = 0 \qquad \text{for all } \mathbf{r}\in S, \tag{1.1}$$

where $\mathbf{n}(\mathbf{r})$ is the unit vector normal to S at $\mathbf{r}$. The basins do not overlap, and their union is $\mathbb{R}^3$ apart from the boundary surfaces, which have zero volume. An integral over $\mathbb{R}^3$ is therefore the sum of the integrals over the basins.

1.2 Basin population and atomic charge

Definition 1.1 — Basin population

$$N(\mathcal{A}) = \int_{\mathcal{A}} \rho(\mathbf{r})\,\mathrm{d}\mathbf{r}, \qquad \sum_{\mathcal{A}} N(\mathcal{A}) = N. \tag{1.2}$$

The population $N(\mathcal{A})$ is the average number of electrons in $\mathcal{A}$. It is dimensionless: the unit a_0^{-3} of ρ cancels against the unit a_0^3 of $\mathrm{d}\mathbf{r}$. The atomic charge of an atom with nuclear charge $Z_{\mathcal{A}}$ is

$$q(\mathcal{A}) = Z_{\mathcal{A}} - N(\mathcal{A}) \qquad [e]. \tag{1.3}$$

Spin-resolved populations are $N^\sigma(\mathcal{A}) = \int_{\mathcal{A}} \rho^\sigma\,\mathrm{d}\mathbf{r}$.

1.3 Basin-integrated pair density

Definition 1.2 — Basin-integrated pair density

$$N_2(\mathcal{A},\mathcal{B}) = \int_{\mathcal{A}} \mathrm{d}\mathbf{r}_1 \int_{\mathcal{B}} \mathrm{d}\mathbf{r}_2\, \rho_2(\mathbf{r}_1,\mathbf{r}_2). \tag{1.4}$$

The quantity $N_2(\mathcal{A},\mathcal{B})$ is the expected number of ordered pairs of distinct electrons with one electron in $\mathcal{A}$ and the other in $\mathcal{B}$. Its unit is $a_0^{-6}\cdot a_0^6$, so N_2 is dimensionless. The normalization of ρ_2 gives $\sum_{\mathcal{A},\mathcal{B}} N_2(\mathcal{A},\mathcal{B}) = N(N-1)$. Integration of Eq. (3.7) of Tome 1, $\int \rho_2 \mathrm{d}\mathbf{r}_2 = (N-1)\rho(\mathbf{r}_1)$, over $\mathcal{A}$ gives

$$\sum_{\mathcal{B}} N_2(\mathcal{A},\mathcal{B}) = (N-1)\,N(\mathcal{A}). \tag{1.5}$$

Relation to the electron-counting operator. Let $\hat{N}_{\mathcal{A}} = \sum_{k=1}^{N} \theta_{\mathcal{A}}(\mathbf{r}_k)$, where $\theta_{\mathcal{A}}(\mathbf{r}) = 1$ inside $\mathcal{A}$ and 0 outside. Then

$$\hat{N}_{\mathcal{A}}\hat{N}_{\mathcal{B}} = \sum_{k} \theta_{\mathcal{A}}(\mathbf{r}_k)\theta_{\mathcal{B}}(\mathbf{r}_k) + \sum_{k\neq l} \theta_{\mathcal{A}}(\mathbf{r}_k)\theta_{\mathcal{B}}(\mathbf{r}_l). \tag{1.6}$$

The first sum is nonzero only for $\mathcal{A} = \mathcal{B}$, since the basins do not overlap, and equals $\hat{N}_{\mathcal{A}}$ in that case. The expectation value of the second sum is $N_2(\mathcal{A}, \mathcal{B})$. Hence

$$\langle \hat{N}_{\mathcal{A}} \hat{N}_{\mathcal{B}} \rangle = N_2(\mathcal{A}, \mathcal{B}) + \delta_{\mathcal{A}\mathcal{B}}\, N(\mathcal{A}). \tag{1.7}$$

2. Population Fluctuations

The population of a basin fluctuates between measurements of the electron positions. The variance and covariance of the populations follow from Eq. (1.7) and the definition of the exchange-correlation density, $\rho_2 = \rho\rho + \gamma_{\mathrm{xc}}$.

2.1 Covariance and variance

Integration of $\rho_2 = \rho(\mathbf{r}_1)\rho(\mathbf{r}_2) + \gamma_{\mathrm{xc}}(\mathbf{r}_1, \mathbf{r}_2)$ over $\mathcal{A} \times \mathcal{B}$ gives

$$N_2(\mathcal{A}, \mathcal{B}) = N(\mathcal{A})N(\mathcal{B}) + \int_{\mathcal{A}} \mathrm{d}\mathbf{r}_1 \int_{\mathcal{B}} \mathrm{d}\mathbf{r}_2 \, \gamma_{\mathrm{xc}}(\mathbf{r}_1, \mathbf{r}_2). \tag{2.1}$$

The covariance $\mathrm{cov}(N_{\mathcal{A}}, N_{\mathcal{B}}) = \langle \hat{N}_{\mathcal{A}} \hat{N}_{\mathcal{B}} \rangle - N(\mathcal{A})N(\mathcal{B})$ is then

$$\mathrm{cov}(N_{\mathcal{A}}, N_{\mathcal{B}}) = \int_{\mathcal{A}} \mathrm{d}\mathbf{r}_1 \int_{\mathcal{B}} \mathrm{d}\mathbf{r}_2 \, \gamma_{\mathrm{xc}}(\mathbf{r}_1, \mathbf{r}_2) + \delta_{\mathcal{A}\mathcal{B}} N(\mathcal{A}). \tag{2.2}$$

For $\mathcal{A} \neq \mathcal{B}$ the covariance is the double integral of γ_{xc}. For $\mathcal{A} = \mathcal{B}$ the variance is

$$\sigma^2(N_{\mathcal{A}}) = N(\mathcal{A}) + \int_{\mathcal{A}} \mathrm{d}\mathbf{r}_1 \int_{\mathcal{A}} \mathrm{d}\mathbf{r}_2 \, \gamma_{\mathrm{xc}}(\mathbf{r}_1, \mathbf{r}_2). \tag{2.3}$$

All three quantities are dimensionless.

2.2 Constraint from the fixed number of electrons

The total number of electrons $\sum_{\mathcal{A}} \hat{N}_{\mathcal{A}} = N$ is a constant, so its variance is zero:

$$\sum_{\mathcal{A}} \sigma^2(N_{\mathcal{A}}) + \sum_{\mathcal{A} \neq \mathcal{B}} \mathrm{cov}(N_{\mathcal{A}}, N_{\mathcal{B}}) = 0. \tag{2.4}$$

Therefore the population fluctuations of different basins are correlated, and the covariances are predominantly negative: an electron found in one basin is absent from the others. Chapter 3 relates these quantities to the localization and delocalization indices.

3. Localization and Delocalization Indices

3.1 Definitions

Definition 3.1 — Localization and delocalization indices

$$\lambda(\mathcal{A}) = -\int_{\mathcal{A}} \mathrm{d}\mathbf{r}_1 \int_{\mathcal{A}} \mathrm{d}\mathbf{r}_2\, \gamma_{\mathrm{xc}}(\mathbf{r}_1,\mathbf{r}_2), \qquad \delta(\mathcal{A},\mathcal{B}) = -2\int_{\mathcal{A}} \mathrm{d}\mathbf{r}_1 \int_{\mathcal{B}} \mathrm{d}\mathbf{r}_2\, \gamma_{\mathrm{xc}}(\mathbf{r}_1,\mathbf{r}_2) \quad (\mathcal{A}\neq\mathcal{B}). \tag{3.1}$$

Both indices are dimensionless. The index $\lambda(\mathcal{A})$ is the number of electrons localized in $\mathcal{A}$: it is the part of the population $N(\mathcal{A})$ that is not exchanged with other basins. The index $\delta(\mathcal{A},\mathcal{B})$ is the number of electron pairs shared between $\mathcal{A}$ and $\mathcal{B}$. The factor 2 accounts for the two ordered contributions $\mathcal{A}\to\mathcal{B}$ and $\mathcal{B}\to\mathcal{A}$, which are equal because γ_{xc} is symmetric.

3.2 Relations to the population fluctuations

Comparison of Eq. (3.1) with Eq. (2.3) and with the covariance gives

$$\sigma^2(N_{\mathcal{A}}) = N(\mathcal{A}) - \lambda(\mathcal{A}), \qquad \mathrm{cov}(N_{\mathcal{A}},N_{\mathcal{B}}) = -\tfrac{1}{2}\,\delta(\mathcal{A},\mathcal{B}) \quad (\mathcal{A}\neq\mathcal{B}). \tag{3.2}$$

The delocalization index is minus twice the covariance of the basin populations. A positive δ corresponds to anticorrelated populations: when $\mathcal{A}$ contains more electrons than its average, $\mathcal{B}$ contains fewer. Since a variance is non-negative, $\lambda(\mathcal{A}) \le N(\mathcal{A})$.

3.3 Sum rules

Integration of $\int \gamma_{\mathrm{xc}}(\mathbf{r}_1,\mathbf{r}_2)\mathrm{d}\mathbf{r}_2 = -\rho(\mathbf{r}_1)$ (Eq. (4.2) of Tome 1) over $\mathbf{r}_1 \in \mathcal{A}$ gives

$$\int_{\mathcal{A}} \mathrm{d}\mathbf{r}_1 \int_{\mathbb{R}^3} \mathrm{d}\mathbf{r}_2\, \gamma_{\mathrm{xc}}(\mathbf{r}_1,\mathbf{r}_2) = -N(\mathcal{A}). \tag{3.3}$$

Separation of the $\mathbf{r}_2$ integral into the basin $\mathcal{A}$ and the other basins yields

$$N(\mathcal{A}) = \lambda(\mathcal{A}) + \frac{1}{2}\sum_{\mathcal{B}\neq\mathcal{A}} \delta(\mathcal{A},\mathcal{B}), \qquad N = \sum_{\mathcal{A}} \lambda(\mathcal{A}) + \frac{1}{2}\sum_{\mathcal{A}}\sum_{\mathcal{B}\neq\mathcal{A}} \delta(\mathcal{A},\mathcal{B}). \tag{3.4}$$

Each electron of $\mathcal{A}$ is counted either as localized in $\mathcal{A}$ or as shared with another basin. The factor $1/2$ avoids counting each shared pair twice. Equation (3.4) is equivalent to Eq. (3.2) and to the constraint of Section 2.2.

Property 3.1 — Summary of relations

$$\begin{aligned} &\sigma^2(N_{\mathcal{A}}) = N(\mathcal{A}) - \lambda(\mathcal{A}) = \tfrac{1}{2}\sum_{\mathcal{B}\neq\mathcal{A}} \delta(\mathcal{A},\mathcal{B}), \\ &\mathrm{cov}(N_{\mathcal{A}},N_{\mathcal{B}}) = -\tfrac{1}{2}\,\delta(\mathcal{A},\mathcal{B}), \qquad 0 \le \lambda(\mathcal{A}) \le N(\mathcal{A}). \end{aligned} \tag{3.5}$$

Interpretation. For a single covalent bond between two equivalent atoms in a closed-shell single-determinant description, $\delta(\mathcal{A}, \mathcal{B})$ is close to one pair of shared electrons. For two basins connected by an ionic interaction, the electrons are localized in the basins and δ is lower. The index quantifies sharing without assigning a bond order by convention.

4. Orbital Expressions

4.1 Atomic overlap matrix

Definition 4.1 — Atomic overlap matrix

For orbitals ϕ_i, the matrix element restricted to basin $\mathcal{A}$ is

$$S_{ij}(\mathcal{A}) = \int_{\mathcal{A}} \phi_i^*(\mathbf{r})\, \phi_j(\mathbf{r})\, \mathrm{d}\mathbf{r}. \tag{4.1}$$

The units are dimensionless: $a_0^{-3/2} \cdot a_0^{-3/2} \cdot a_0^3 = 1$. The matrix $\mathbf{S}(\mathcal{A})$ is Hermitian and positive semidefinite, because for any coefficients c_i,

$$\sum_{ij} c_i^* S_{ij}(\mathcal{A}) c_j = \int_{\mathcal{A}} \Big|\sum_j c_j \phi_j\Big|^2 \mathrm{d}\mathbf{r} \;\geq 0. \tag{4.2}$$

Orthonormality of the orbitals over $\mathbb{R}^3$ gives $\sum_{\mathcal{A}} S_{ij}(\mathcal{A}) = \delta_{ij}$.

4.2 Natural-orbital form

For a one-electron reduced density matrix with natural orbitals ϕ_i and occupation numbers n_i in the spin-summed convention,

$$\gamma(\mathbf{r}_1, \mathbf{r}_2) = \sum_i n_i\, \phi_i(\mathbf{r}_1)\phi_i^*(\mathbf{r}_2), \qquad N(\mathcal{A}) = \sum_i n_i S_{ii}(\mathcal{A}). \tag{4.3}$$

The Ángyán–Loos–Mayer (ALM) approximation reconstructs the exchange-correlation density from γ,

$$\gamma_{\mathrm{xc}}^{\mathrm{ALM}}(\mathbf{r}_1, \mathbf{r}_2) = -\tfrac{1}{2}\, |\gamma(\mathbf{r}_1, \mathbf{r}_2)|^2. \tag{4.4}$$

Expansion of the modulus,

$$|\gamma(\mathbf{r}_1, \mathbf{r}_2)|^2 = \sum_{ij} n_i n_j\, \phi_i(\mathbf{r}_1)\phi_j^*(\mathbf{r}_1)\, \phi_i^*(\mathbf{r}_2)\phi_j(\mathbf{r}_2), \tag{4.5}$$

and integration over $\mathcal{A} \times \mathcal{B}$, with $\int_{\mathcal{A}} \phi_i \phi_j^* \mathrm{d}\mathbf{r} = S_{ji}(\mathcal{A})$ and $\int_{\mathcal{B}} \phi_i^* \phi_j \mathrm{d}\mathbf{r} = S_{ij}(\mathcal{B})$, give

$$\delta^{\mathrm{ALM}}(\mathcal{A}, \mathcal{B}) = \sum_{ij} n_i n_j\, S_{ij}(\mathcal{A})\, S_{ji}(\mathcal{B}), \qquad \lambda^{\mathrm{ALM}}(\mathcal{A}) = \frac{1}{2} \sum_{ij} n_i n_j\, S_{ij}(\mathcal{A})\, S_{ji}(\mathcal{A}). \tag{4.6}$$

The expression is bilinear in the products $n_i n_j$ and in the matrices $\mathbf{S}(\mathcal{A})$ and $\mathbf{S}(\mathcal{B})$. Fractional occupation numbers carry the effect of electron correlation into δ.

4.3 Single determinant

For a closed-shell determinant, $n_i = 2$ for the $N/2$ occupied orbitals and $n_i = 0$ for the others. Equation (4.6) reduces to

$$\delta^{\mathrm{HF}}(\mathcal{A}, \mathcal{B}) = 4 \sum_{i,j}^{N/2} S_{ij}(\mathcal{A})\, S_{ji}(\mathcal{B}) = 4\, \mathrm{Tr}[\mathbf{S}(\mathcal{A})\mathbf{S}(\mathcal{B})], \qquad \lambda^{\mathrm{HF}}(\mathcal{A}) = 2\, \mathrm{Tr}[\mathbf{S}(\mathcal{A})^2]. \tag{4.7}$$

For an open-shell determinant, the spin blocks contribute separately,

$$\delta^{\mathrm{HF}}(\mathcal{A},\mathcal{B}) = 2\sum_{\sigma}\sum_{i,j}^{N^{\sigma}} S^{\sigma}_{ij}(\mathcal{A})\, S^{\sigma}_{ji}(\mathcal{B}). \tag{4.8}$$

For $N^{\uparrow} = N^{\downarrow} = N/2$ and identical spatial orbitals, this expression equals Eq. (4.7).

Property 4.1 — Non-negativity

For a closed-shell determinant, $\delta^{\mathrm{HF}}(\mathcal{A},\mathcal{B}) \geq 0$. The trace of a product of two positive semidefinite matrices is non-negative, and $\mathbf{S}(\mathcal{A})$ and $\mathbf{S}(\mathcal{B})$ are positive semidefinite. The same argument applies to δ^{ALM} for $n_i \geq 0$, with $\mathbf{S}(\mathcal{A})$ replaced by $\mathbf{n}^{1/2}\mathbf{S}(\mathcal{A})\mathbf{n}^{1/2}$, where $\mathbf{n} = \mathrm{diag}(n_i)$.

Example 4.1 — Hydrogen molecule, single determinant

One doubly occupied orbital σ_g is symmetric with respect to the two equivalent nuclei. The orbital is normalized, and $S(\mathcal{A}) + S(\mathcal{B}) = 1$ with $S(\mathcal{A}) = S(\mathcal{B})$ by symmetry, so $S(\mathcal{A}) = S(\mathcal{B}) = \frac{1}{2}$. Then

$$\delta(\mathcal{A},\mathcal{B}) = 4 \cdot \tfrac{1}{2} \cdot \tfrac{1}{2} = 1, \qquad \lambda(\mathcal{A}) = 2 \cdot \tfrac{1}{4} = \tfrac{1}{2}, \qquad N(\mathcal{A}) = 2 \cdot \tfrac{1}{2} = 1. \tag{4.9}$$

The sum rule is satisfied: $N(\mathcal{A}) = \lambda(\mathcal{A}) + \frac{1}{2}\delta(\mathcal{A},\mathcal{B}) = \frac{1}{2} + \frac{1}{2} = 1$. The population variance is $\sigma^2(N_{\mathcal{A}}) = N(\mathcal{A}) - \lambda(\mathcal{A}) = \frac{1}{2}$ and $\mathrm{cov}(N_{\mathcal{A}}, N_{\mathcal{B}}) = -\frac{1}{2}$.

5. Response to the Number of Electrons

5.1 Electron-number derivatives

The delocalization index depends on the electron number N through the occupation numbers and the orbitals. For a fractional electron number $N = N_0 + \omega$ with $0 \leq \omega \leq 1$, the state is an ensemble of the N_0-electron and $(N_0 + 1)$-electron states (Perdew, Parr, Levy and Balduz, PPLB). Quantities defined for the ensemble can have different derivatives on the two sides of an integer N_0. The one-sided derivatives are

$$\left(\frac{\partial\delta(\mathcal{A},\mathcal{B})}{\partial N}\right)_- = \lim_{\Delta N\to 0^-} \frac{\delta(\mathcal{A},\mathcal{B};N_0+\Delta N)-\delta(\mathcal{A},\mathcal{B};N_0)}{\Delta N}, \tag{5.1}$$

$$\left(\frac{\partial\delta(\mathcal{A},\mathcal{B})}{\partial N}\right)_+ = \lim_{\Delta N\to 0^+} \frac{\delta(\mathcal{A},\mathcal{B};N_0+\Delta N)-\delta(\mathcal{A},\mathcal{B};N_0)}{\Delta N}. \tag{5.2}$$

The derivative with $\Delta N < 0$ describes electron removal. The derivative with $\Delta N > 0$ describes electron addition.

5.2 Dual Delocalization Descriptor

Definition 5.1 — Dual Delocalization Descriptor

$$\Delta\delta(\mathcal{A},\mathcal{B}) = \left(\frac{\partial\delta(\mathcal{A},\mathcal{B})}{\partial N}\right)_+ - \left(\frac{\partial\delta(\mathcal{A},\mathcal{B})}{\partial N}\right)_- . \tag{5.3}$$

The delocalization index δ is dimensionless. The derivative with respect to N, counted in electrons, has the unit of an inverse electron number, and so has $\Delta\delta$.

Sign convention. A positive $\Delta\delta(\mathcal{A},\mathcal{B})$ indicates a net increase of the electron sharing between $\mathcal{A}$ and $\mathcal{B}$ in the response to electron addition and removal. A negative value indicates a net decrease. The value depends on the relative weight of the electron-attachment and electron-removal channels.

5.3 Differentiability and the neutral reference state

The derivatives require a differentiable dependence of δ on N. The bilinear ALM form, Eq. (4.6), gives finite one-sided derivatives and separates inter-orbital contributions ($i \neq j$) from self-delocalization contributions ($i = j$). The square-root (Müller-type) functional of the occupation numbers has no well-defined one-sided derivative with respect to N in the PPLB framework. The analytical expressions evaluate $\Delta\delta$ from the ground-state wave function of the neutral species, without calculations on the ionic states. The derivation and the numerical validation on aromatic systems are given in Ref. [6].

A. Exercises

1. Show that $\sum_{\mathcal{A}} \hat{N}_{\mathcal{A}} = N$ implies $\sum_{\mathcal{A}} \sigma^2(N_{\mathcal{A}}) + \sum_{\mathcal{A}\neq\mathcal{B}} \mathrm{cov}(N_{\mathcal{A}}, N_{\mathcal{B}}) = 0$, and derive from this identity the total sum rule of Eq. (3.4).

2. For the hydrogen molecule of Example 4.1, compute $\sigma^2(N_{\mathcal{A}})$ and verify the identity of Exercise 1.

3. Show that $\mathbf{S}(\mathcal{A})$ is positive semidefinite and that $\delta^{\mathrm{HF}}(\mathcal{A}, \mathcal{B}) \geq 0$.

4. A closed-shell system has $N = 10$ electrons. Give $\sum_{\mathcal{A},\mathcal{B}} N_2(\mathcal{A}, \mathcal{B})$.

5. Give the unit of $\partial\delta/\partial N$ when N is counted in electrons.

B. Solutions

1. A constant has zero variance, so $0 = \mathrm{var}(\sum_{\mathcal{A}} \hat{N}_{\mathcal{A}}) = \sum_{\mathcal{A}} \sigma^2(N_{\mathcal{A}}) + \sum_{\mathcal{A}\neq\mathcal{B}} \mathrm{cov}(N_{\mathcal{A}}, N_{\mathcal{B}})$. Substitution of Eq. (3.2) gives $\sum_{\mathcal{A}}[N(\mathcal{A}) - \lambda(\mathcal{A})] - \frac{1}{2}\sum_{\mathcal{A}\neq\mathcal{B}} \delta(\mathcal{A},\mathcal{B}) = 0$. Since $\sum_{\mathcal{A}} N(\mathcal{A}) = N$, this is $N = \sum_{\mathcal{A}} \lambda(\mathcal{A}) + \frac{1}{2}\sum_{\mathcal{A}\neq\mathcal{B}} \delta(\mathcal{A},\mathcal{B})$.

2. $\sigma^2(N_{\mathcal{A}}) = \sigma^2(N_{\mathcal{B}}) = \frac{1}{2}$ and $\mathrm{cov}(N_{\mathcal{A}}, N_{\mathcal{B}}) = -\frac{1}{2}$. The identity reads $\frac{1}{2} + \frac{1}{2} + 2\cdot(-\frac{1}{2}) = 0$.

3. The form $\sum c_i^* S_{ij} c_j = \int_{\mathcal{A}} |\sum_j c_j \phi_j|^2 \mathrm{d}\mathbf{r} \geq 0$ shows that $\mathbf{S}(\mathcal{A})$ is positive semidefinite. For positive semidefinite $\mathbf{A}$ and $\mathbf{B}$, $\mathrm{Tr}(\mathbf{AB}) = \mathrm{Tr}(\mathbf{A}^{1/2}\mathbf{B}\mathbf{A}^{1/2}) \geq 0$. Thus $\delta^{\mathrm{HF}} = 4\,\mathrm{Tr}[\mathbf{S}(\mathcal{A})\mathbf{S}(\mathcal{B})] \geq 0$.

4. $\sum_{\mathcal{A},\mathcal{B}} N_2(\mathcal{A},\mathcal{B}) = N(N-1) = 90$.

5. The delocalization index is dimensionless, so $\partial\delta/\partial N$ has the unit N^{-1} (inverse electron number).

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